

Roomard

The AI guest-memory engine.

For boutique hotels. Capture every preference. Brief every shift. Prepare every room.

The problem

Boutique hotels run on memory that walks out the door.

A returning guest's pillow preference, the allergy from last stay, the complaint that was never closed - all of it lives in one supervisor's head, a paper card in a drawer, or a concierge inbox nobody else can see.

- Staff turnover erases the guest relationship overnight.
- Chains have \$200k CRMs; independents have sticky notes.
- The data exists - handwritten cards, emails, reviews, PMS records - but it is never unified, never surfaced at the front desk, never auditable.

The solution

One memory engine. Three moments that matter.

CAPTURE

Handwritten check-in cards, concierge emails, PMS records and reviews flow into one guest profile. PaddleOCR-VL + ERNIE 4.5 turn paper into structured, source-attributed preferences.

BRIEF

Every morning, an AI arrival brief prioritises today's guests - VIP, repeat, prior-issue - with one-tap evidence drill-down. Degrades gracefully when AI is unavailable.

PREPARE

Mid-conversation lookup, D-1 housekeeping prep cards, and a complaint-trajectory flag that catches a worsening guest before the next stay. Every action audited.

The MVP wedge

Eight use cases. Seven built. One honest stub.

UC-07	Daily arrival brief	90%
UC-01	Handwritten card capture (PaddleOCR-VL)	85%
UC-08	Mid-conversation guest lookup	80%
UC-09	Housekeeping prep card (D-1)	80%
UC-24a	Mews inbound PMS sync	80%
UC-23	Confidence / exception queue	75%
UC-11	Complaint trajectory (ERNIE X1) - bonus	75%
UC-25	Review polling (pipeline real, APIs stubbed)	70%
UC-29	SSO (SAML/OIDC) - honest 501	5%

The AI stack

Sponsor models, routed through one gateway.

PaddleOCR-VL

Layout-aware OCR

Handwritten check-in cards → text + structure

ERNIE 4.5

Entity extraction

Card/email/review text → structured preferences

ERNIE X1

Reasoning

Complaint-trajectory flag (3-issue rule, UC-11)

Qianfan MaaS

Inference gateway

Single path to models; per-tenant caps; every call logged

AI degrades gracefully everywhere: when a model is unavailable the brief still renders, the prep card falls back to a plain card, and the trajectory flag falls back to the rule-based threshold.

Live demo · 4 stages

What you will see in the walkthrough.

1

DAILY ARRIVAL BRIEF (UC-07)

Front-desk manager signs in and lands on the morning brief: prioritised arrivals, AI-written notes, one-tap evidence.

2

GUEST LOOKUP + TRAJECTORY (UC-08 / UC-11)

Search a guest mid-conversation: 3-bullet preferences, last-stay summary, 'say this' line, complaint-trajectory flag.

3

HANDWRITTEN CARD CAPTURE (UC-01)

Photograph a paper check-in card; PaddleOCR-VL + ERNIE 4.5 extract fields; low confidence routes to the queue.

4

EXCEPTION QUEUE + PREP CARDS (UC-23 / UC-09)

Clear the confidence queue; review tomorrow's housekeeping prep cards; two-tap completion; every action audited.

Proof, not promises

Engineering rigour you can re-run.

356

unit tests pass

12

DB integration tests (live PG)

87.5 %

mean module coverage

39

findings fixed · 0 open

3

production bugs caught by tests

0 / 0

lint errors / warnings

78

traceable commits on main

RLS

tenant isolation proven live

Reproducible: `pnpm -r test` · `pnpm -r run test:coverage` · DB integration gated on `DATABASE_URL`

Architecture

A typed monorepo. Ten services. One edge.

FRONTEND

React 18 · TanStack Router · Vite · Tailwind (PWA)

Offline capture queue · served on :8180

EDGE + SERVICES

API gateway + 10 Fastify services (auth, guest, capture, brief, exception, audit, ingest, tenant, ai-gateway)

JWT + RBAC at the edge · gateway on :3100

DATA + AI

Postgres 16 (RLS-enforced) · Redis · MinIO · Qianfan MaaS

Append-only audit · per-tenant row-level security

Why this is trustworthy

Multi-tenant and audit-grade from line one.

Row-level security, proven live

RLS is forced on every tenant table; a dedicated non-superuser app role makes isolation real - demonstrated against the live database, not just claimed.

Append-only audit chain

Every event is hash-chained; the verify endpoint re-derives the chain in SQL so tampering is detectable.

Tests find real bugs

Three production bugs (a tenant-context failure, an audit schema drift, a dead error-forwarder) were caught by chasing real coverage - not padding.

Graceful degradation

Every AI path has a non-AI fallback; the product stays useful when a model is down.

What ships next

From wedge to platform.

Close the wedge

- Real review-API adapters (TripAdvisor / Booking / Google)
- Outbound PMS sync (UC-24b) + a second PMS connector
- SSO via a real IdP test tenant (replace the 501)
- Push-notification delivery for briefs + GM flags
- Benchmark corpora to verify the OCR / ranking AC bars

Platform

- GDPR subject-access + erasure (EU legal gate)
- Native mobile (PWA → Capacitor) for housekeeping
- Multi-language UI (EN / IT / ES / ZH)
- Long-horizon guest memory across properties
- Lock the live stack on the non-superuser role end-to-end

Team + ask

Built for the Baidu Build with MeDo hackathon.

Founder + lead engineer · product, architecture, full-stack build, demo, verification

Generated under the AT-Hack0019 sprint with Claude as paired engineer.

What we are asking for

- Qianfan production-tier access for the live multilingual rollout.
- A boutique-hotel design partner (10-50 keys) for a 4-week pilot.
- A Mews / Cloudbeds partner connection to validate the PMS sync at scale.
- Time to demo the capture → brief → prepare loop with a real property's data.

Roomard

Every preference remembered.

Capture. Brief. Prepare. Multi-tenant, RLS-enforced, audit-grade by default.

github.com/vsenthil7/roomard

356 tests · 12 DB integration · 87.5% mean coverage · 39 findings fixed · 0 open

Thank you.